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Dynamic matching strategy for college students' innovative training projects based on reinforcement learning optimization

Xiao Ju¹ and Yumo Ding^{2*}

*Correspondence:

Yumo Ding
dingym1998@outlook.com
¹Liaodong University,
Dandong 118003, Liaoning, China
²The National University of
Malaysia, 43600 Bangi, Selangor,
Malaysia

Abstract

Matching college students with appropriate innovative training projects is a challenging task that often relies on static assignment techniques, which overlook individual interests, skills, and learning styles. Traditional methods lead to mismatched assignments, decreased engagement, and lower innovation output. This research presents an intelligent, dynamic matching algorithm that maximizes the allocation of students to innovative training projects, utilizing techniques based on reinforcement learning (RL) to analyze interactions and continually optimize assignment decisions. The data was collected, which involves the profiles of students, their interests and skills rating, and project metadata. The data is pre-processed to normalize and encode categorical features. The extraction of features, dimensionality reduction, and significant matching signs are obtained by the Principal Component Analysis (PCA). The Q-Learning Algorithm Tuned Dueling Deep Q-Network (QLA-D2QNet) was developed to dynamically learn optimal matching policies through interaction with the environment and reward feedback. QLA is used to learn optimal assignment strategies by trial and error. D2QNet separates value and advantage functions to enhance policy learning stability. The model constantly adjusts matching policies based on feedback for project success and student satisfaction. The experimental results indicate that the QLA-D2QNet significantly outperforms traditional manual approaches. The best results include a Skill-Interest Fit of 89.10%, a Project Completion Rate of 94.00%, and a Skill Improvement Score of 31.40%. The suggested QLA-D2QNet model provides a scalable, flexible, and successful technique for dynamically matching students to training projects, resulting in dramatically improved educational outcomes in creative learning environments.

Keywords College students, training projects, innovative training projects, assignment decisions, Q-Learning Algorithm Tuned Dueling Deep Q-Network (QLA-D2QNet)

1 Introduction

The concept of advanced training programs has recently become a crucial component of higher education, allowing students to develop their creativity, problem-solving skills, and workplace skills within the framework of real-life projects [1]. These programs are



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typically envisaged as group or individual projects in any of numerous fields that require cooperation and a cautious balancing of student capabilities with the project goals [2]. This is the most delicate task in this situation because it involves the successful and careful assignment of students to relevant projects [3]. A perfect matching process must extend past credentials and also interests, skills, learning styles, and development goals [4]. Many systems of assigning students statically do not consider factors, which brings about ill-matched assignments, a lack of student participation, and overall poor project deliverables. In turn, it has resulted in the necessity to design dynamic and data-driven models of decision-making to improve the efficacy of student-project matching [5]. Figure 1 illustrates an RL-driven dynamic student-project matching framework.

To solve the matching problem, conventional methods employed rule-based approaches, machine learning (ML), and deep learning (DL) methodologies. The ML algorithms, for example, support vector machines (SVM) or decision trees (DT), were applied to predict the student preference or success chance from past features [6–7]. Latent correlations involving students and project attributes can be found using DL models like neural networks or autoencoders. Their dependence on pre-labeled data and static nature, however, restricts them. The majority of models only make predictions once and are not flexible enough to adjust to shifting student preferences or project specifications [8–9]. A failure to be interpreted renders it less suitable for dynamic academic environments with frequent feedback and contextual changes [10–11]. This research offers a dynamic, reinforcement learning (RL)-driven matching method that learns and improves student-project assignments over time to get around these limitations. To overcome the problem as a sequential decision-making process, a QLA-D2QNet is specifically introduced. Combining the benefits of dueling DQN and Q-learning, QLA-D2QNet is a dynamic, feedback-driven approach. It changes its policies, solicits feedback, and interacts with the environment to make better decisions. Strong policy learning and more accurate and tailored matching are two ways that the

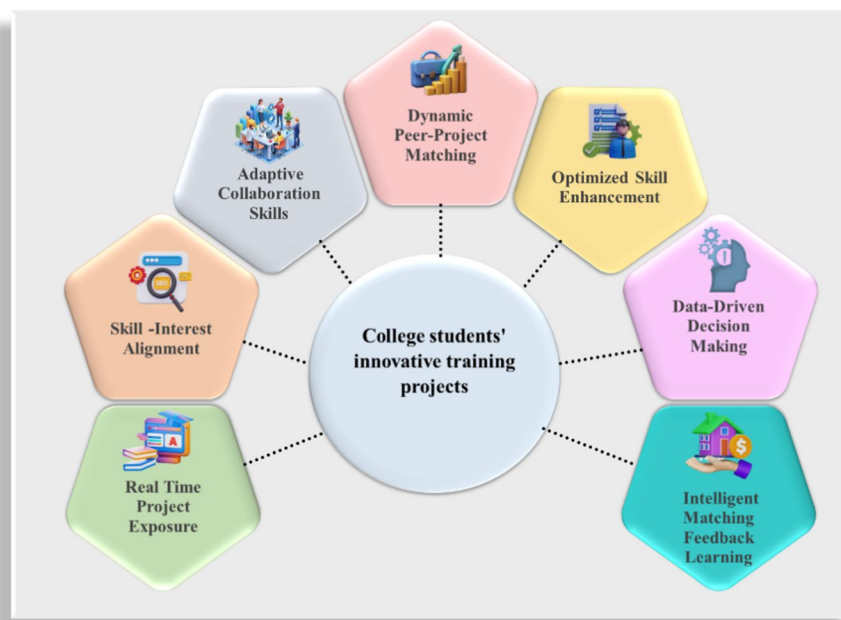


Fig. 1 Factors of college students' innovative training projects

dynamic method enhances innovation outcomes in educational settings. The following is the section on contributions:

- Creating an in-depth student-project dataset

As the basis for intelligent matching, a structured dataset is created that includes student profiles, project details, interests, and talents.

- Data preprocessing for enhanced model input

Data that has been collected is cleaned, normalized, and encoded. Numerical characteristics are scaled, and categorical variables are one-hot encoded. These preparation procedures guarantee uniformity, lessen bias, and get the data ready for efficient RL model training and learning.

- Application of PCA for feature extraction

PCA improves model interpretability and computing efficiency by highlighting important matching characteristics and reducing dimensionality.

- Construction of the QLA-D2QNet

Q-Learning and Dueling DQN architecture are both incorporated within the QLA-D2QNet model. The decoupling of value and benefits functions enhances learning stability and allows it to adjust policies more accurately to adapt students to the most advanced training projects.

- Assessment of experiments and comparison

Experimental validation of the suggested model is conducted against conventional methods. The usefulness of QLA-D2QNet in actual educational settings is confirmed by the results, which demonstrate significant gains in training skills, student involvement, and re-application rates.

The research is structured into the following parts: Part 2 reviews related work on student-project matching methods and reinforcement learning applications in education. Part 3 introduces the proposed QLA-D2QNet framework designed to dynamically optimize student assignments based on continuous feedback. Part 4 presents the experimental results and discussion, highlighting the model's performance compared to traditional methods. Finally, Part 5 summarizes the research and outlines potential directions for future development and real-world implementation.

2 Related work

A dynamic approach [12] has been developed to improve vocational education by adapting to students' learning patterns during hands-on training. The system helped instructors make more effective decisions through Markov decision-making and Q-learning algorithms. Results indicate that students who received their dynamic support performed better in exams and felt more confident and motivated to learn.

The improvement of online learning for students based on their choice of suitable courses and activities based on their needs was the key intention of this research. Markov decision processes and RL were used to recommend personalized learning paths [13]. Results show better learning outcomes and increased engagement, highlighting the method's effectiveness in guiding students through their online educational journey. Innovative

approaches such as IDEATION and DEETECHTIVE could support the growth of technology-based enterprises in universities of higher learning, as examined in [14]. The findings demonstrate that both methods were effective in fostering creativity, enhancing students' innovative thinking, and fortifying ties between academics and business.

The impact of student engagement and Project-Based online instruction (PBOI) on academic achievement in entrepreneurial education was examined in [15]. According to the results, PBOI and higher levels of student participation enhanced academic achievement, encouraged the growth of entrepreneurial skills, and produced a more pleasurable and productive educational atmosphere.

The effects of project-based learning (PBL), team cohesiveness, and flipped learning on student participation and educational effectiveness were investigated. Utilizing data from 258 business students and multivariate analysis, the research [16] verified that team cohesion and PBL both had a positive effect on participation and student retention. Participation decreased with flipped instruction, though educational performance improved. Student participation also had a positive effect on learning outcomes.

To enhance individualized learning, Q-Learning was used to modify instructional materials according to student performance and interactions, as examined in [17]. A reinforcement learning model was constructed in which learner success determines rewards, and choices for instructional content operate as actions. When integrated into platforms such as Moodle, the approach discovered content tactics that work, leading to better learning outcomes and more student engagement in a variety of educational scenarios.

AI-based tasks were explored in [18] to determine their impact on enhancing creativity in undergraduate students. A support vector machine (SVM) model was used to assess performance. Results showed clear improvements in critical and creative thinking, and the SVM accurately predicted student outcomes, demonstrating its effectiveness in evaluating and supporting innovation. An effective student grouping strategy for educational contexts, persona-based Student Grouping (PSG), was suggested in [19], which assigned personas to students depending on behavior information using unsupervised clustering. Experimental results from MOOC datasets reveal that PSG beats manually and arbitrarily sorting in terms of efficiency and efficacy.

Students' poor Course Learning Outcomes (CLOs) were improved by proposing individualized online resources. A dataset based on the outcome-based learning (OBL) paradigm was generated [20], and a student-to-CLO matrix was constructed. Using Q-learning, the method determined ideal learning routes, and a mobile app suggested focused online resources, successfully improving students' inadequate CLOs. Table 1 shows relevant works supporting dynamic matching using reinforcement learning in education.

2.1 Problem statement

Student allocation to suitable training projects remains a persistent issue, particularly in active training environments where competencies, learning styles, and interests tend to be very diverse. Static or rule-based traditional assignments are not able to address special needs, resulting in low motivational levels and suboptimal outcomes. RL, particularly Q-learning, provides a viable method for optimizing customized decision-making through interaction and feedback. Prior research has effectively utilized RL to improve customized learning routes [13] and instructional material delivery [17], indicating a high likelihood of adopting comparable tactics to dynamic project-student matching. The QLA-D2QNet

Table 1 Summary of the relevant works

References	Method	Result	Advantage	Limitations
[21]	Comparative analysis using pre- and post-tests with online questionnaires; content analysis for qualitative data	AI-enhanced teaching and learning improved knowledge mastery, ability development, and emotional engagement	Promotes personalized learning, intelligent lesson prep, and real-time feedback	Context-specific; limited to AI platforms like Watson and Knewton; generalizability may vary
[22]	Mixed system dynamics modeling with simulation via Vensim, based on interviews and investigations	Team capability, motivation, and organizational factors significantly affect knowledge transfer in mega projects	Offers a simulation-based strategy for innovation coordination in complex projects	Model accuracy was limited by the difficulty in gathering precise historical data
[23]	MARLFS (Multi-agent RL-based Feature Selection) for software effort estimation	Improved prediction accuracy over traditional estimation techniques	Integrates project management data with RL-driven feature selection	Depending on the availability and quality of software project data, it might lack interpretability
[24]	Developed a Project-based Learning model integrated with the Think–Pair–Share strategy; experimental study with the control group	Significant improvement in students' creative thinking across all domains (fluency, flexibility, originality, elaboration)	Promotes collaborative learning and creativity development in education students	Research limited to one university in Thailand; may not generalize across contexts
[25]	Meta-analysis with fuzzy inference system to map course attainment to program outcomes	Demonstrated ability to link assessment to learning outcomes using fuzzy logic	Provides a mathematical tool to evaluate individual and group outcomes realistically	Preliminary; limited to engineering courses; lacks long-term validation
[26]	Heuristic approach to SPA-ST (student-project allocation with lecturer preferences over students, with ties)	Achieved near-optimal stable matchings efficiently across random and large dataset experiments	Handles preference ties; computationally efficient; suitable for large student cohorts	Still static; limited to preference lists; does not consider qualitative factors (skills, motivation, outcomes)
[27]	Enhanced collaborative filtering with time-decay weighting + interest migration to recommend projects	Improved recommendation accuracy and coverage compared to standard CF baselines	Handles evolving interests; scalable for large project pools; good personalization	No stability guarantee; doesn't manage capacity constraints; no reinforcement/feedback loop for continuous improvement

that goes beyond static matching by learning dynamic project assignment was overcome by individualized state-action learning and decision improvements.

3 Methodology

The initial step in the proposed process involves collecting data on projects and students, including interests, skills, and project requirements. During preprocessing, while numerical values are normalized and categorical variables are encoded, PCA is applied to reduce dimensionality and highlight key features. The optimal matching strategies are learned using QLA-D2QNet, which enhances learning stability by separating value and advantage functions. Figure 2 illustrates the RL-based student allocation process flow.

3.1 Dataset

The dataset used in this research is sourced from Kaggle [<https://www.kaggle.com/datasets/ziya07/dynamic-student-project-matching-dataset/data>]. It includes 500 creative training

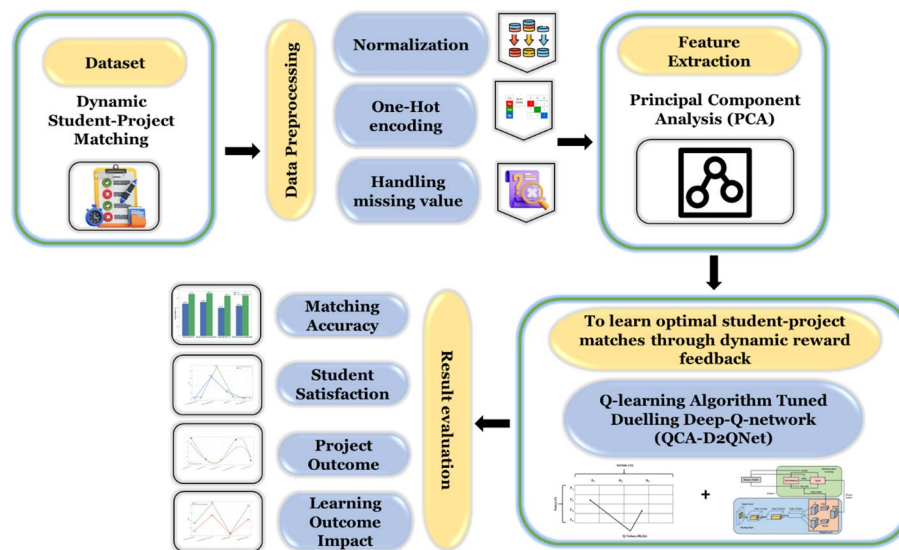


Fig. 2 Student allocation to innovative training projects of QLA-D2QNet

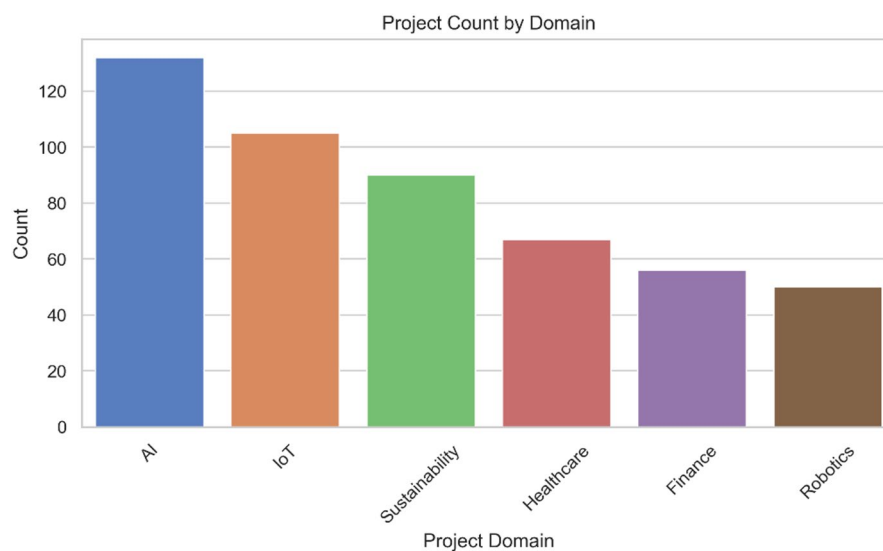


Fig. 3 Distribution of training projects across different domains

projects and 2,000 student records gathered from various colleges between 2021 and 2023. While each project has prerequisites, domain, expected objectives, and complexity level, each student record contains demographics, interests, talents, learning styles, and academic achievement. To ensure equitable representation in training, oversampling is used and stratified batch assignment during preprocessing to address the dataset's imbalance, which is especially evident in the distribution of projects across domains. The distribution of projects across domains and the associated student allocation are shown in Fig. 3.

3.2 Data preprocessing

Project and student data are prepared for model training through data preparation. Normalization is applied to numerical characteristics, such as skill ratings, and numerical values are encoded for categorical data, such as department or project type. To guarantee

completeness, simple imputation techniques are used to handle missing data. Below, the preprocessing procedures are covered in further detail to help understand each step and its significance.

3.2.1 Handling missing values

To ensure data integrity and model performance, missing values must be handled correctly throughout the preprocessing stage. Missing entries are addressed using suitable imputation approaches to ensure consistency, eliminate bias, and improve the reliability of RL-based matching judgments. To create dependable RL models in the dynamic combination across college students and creative teaching initiatives, data completeness is essential. However, mistakes in data recording, incomplete student profiles, or missed project preferences, and real-world educational statistics frequently include missing information.

Imputation methods are used to fill in partial records to overcome this problem. By using this method, the dataset is guaranteed to be uniform and suitable for the optimization phases of RL. The severity is determined as follows, taking into account the amount of missing data: Less than 1% of records have missing data, which are classified as “trivial,” 1–5% as “manageable,” 5–15% as “sophisticated,” and more than 15% as “severe.” The matching approach uses imputation techniques like mean/mode replacement or complex estimates to maintain data integrity and usefulness during policy learning.

3.2.2 Normalization

To guarantee uniform scaling across various student and project attributes, particularly those with varied units or ranges, Z-score normalization is used during data preparation. Exceptions can skew acquisition in RL-based pairing models, but this statistical method effectively mitigates their effects. For each attribute, the mean (μ) and standard deviation (σ) are computed, and each value (v) is transduced through the Eq. (1):

$$\nu' = \frac{\nu - \mu}{\sigma} \quad (1)$$

By normalizing input characteristics, the RL model produces a standardized distribution with a zero center. This prevents results from being distorted by scale discrepancies or extreme values and enables the model to learn the best dynamic project-student matching method effectively.

3.2.3 One-hot encoding (OHE)

For RL algorithms to handle categorical data like project kinds, student interests, or skill categories efficiently, they must be computed in the suggested dynamic matching technique for college students' creative education projects. OHE is a popular target-agnostic encoding technique that creates sparse binary vectors from category variables. Except for a single one at the index that corresponds to the particular category, every vector is made up entirely of zeros. The following is the Eq. (2) representation of the attribute value y_j from a categorical variable with M potential types:

$$y_j = \underbrace{[0, \dots, 1^j, \dots, 0]}_M \quad (2)$$

This vector is usually M in size (one category is excluded to prevent convergence). This work creates M columns for each class feature by applying the Categories Encoders module's default OHE setup. As shown in Fig. 4, the reward distribution highlights the influence of skill–interest alignment, domain relevance, and learning style fit in shaping the optimization outcome.

3.3 Feature extraction of PCA

PCA is employed in feature extraction to minimize data dimensionality, remove redundancy, and bring out the important characteristics. It increases learning efficiency as well as makes the reinforcement learning model concentrate on the most important matching indicators. High-dimensional data containing student profiles, project criteria, and achievement indicators might impede effective computing and learning in the context of adaptive matching between college students and creative training projects. To minimize the dimensionality of such data while maintaining its important variance, PCA is utilized. By removing duplication and noise and emphasizing the most useful traits, this transformation streamlines the RL technique. For feature extraction, PCA is selected because it effectively lowers dimensionality while maintaining the most significant variation in project and student data. Unlike LDA or t-SNE, PCA is appropriate for reinforcement learning tasks because it does not require class labels and preserves the global data structure. By concentrating on the most informative elements, QLA-D2QNet can increase matching accuracy and learning stability.

PCA projects the primary data linearly onto a lower-dimensional domain made up of principal components, which are uncorrelated orthogonal matrices. These elements are obtained using the dataset's covariance matrix's eigenvectors, guaranteeing that every additional dimension preserves the most variation possible. First, PCA calculates the mean vector for a dataset with l observations and w components in Eq. (3):

$$\mu = \frac{1}{l} \sum_{j=1}^l w_1 \quad (3)$$

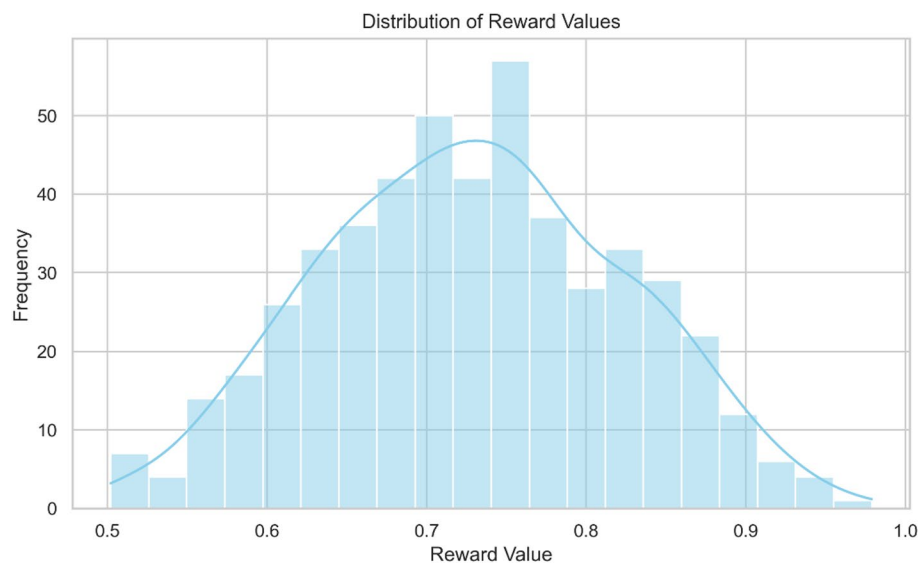


Fig. 4 Distribution of optimized reward values for project matching data feature

where $w_1, w_2, \dots, w_l \in \mathbb{R}^n$. The final projection matrix is obtained by median standardization, covariance matrix calculation, and eigen compression.

The next stage in PCA is to compute the expected covariance matrix T , which illustrates that the dataset's variables differ from one another, after determining the mean vector given by Eq. (4):

$$T = \frac{1}{l} \sum_{j=1}^l (w_j - \mu)(w_j - \mu)^s \quad (4)$$

Finding the directions of highest variation requires the use of this matrix. The eigenvalues $\lambda_1, \geq \lambda_2 \geq \dots \geq \lambda_l \geq 0$ and their matching eigenvectors are produced by conducting an eigen decomposition on T . The primary axes of the numerically distributed data are represented by these eigenvectors. Each major component is calculated using Eq. (5) and represents a continuous composition of the original factors.

$$\begin{aligned} z_1 &= b_{11}w_1 + b_{12}w_2 + b_{1l}w_l \\ z_2 &= b_{21}w_1 + b_{22}w_2 + b_{2l}w_l \\ &\vdots \\ z_l &= b_{l1}w_1 + b_{l2}w_2 + b_{ll}w_l \end{aligned} \quad (5)$$

z_1 captures the largest variation in the original data, after z_2 , and others. These components, $z_1, z_2, \dots, z_l$, are essentially uncorrelated and are placed in decreasing order of the variance.

A small number of primary components frequently account for the bulk of the variance in real-project datasets, particularly ones that include multifaceted student profiles and project features. Considerable dimensionality reduction without considerable information loss is made possible by this characteristic. By choosing the top n components, the percentage of total variance kept is as follows in Eq. (6):

$$\gamma l = \frac{\lambda_1 + \lambda_2 + \dots + \lambda_n}{\lambda_1 + \lambda_2 + \dots + \lambda_n + \dots + \lambda_1} \geq 80\% \quad (6)$$

The retained percentage of the dataset's initial variance is denoted by γl in this case. The majority of the crucial data structure is maintained when components are chosen, so that $\gamma l \geq 80\%$.

The dynamic matched technique simplifies the feature space by keeping just the primary components that account for at least 80% of the total variance. Figure 5 presents the PCA scatter plot, where clusters indicate patterns of student–project outcomes, demonstrating how PCA highlights separability between successful and less successful matches.

3.4 QLA-D2QNet

The QLA-D2QNet is used in the suggested technique to dynamically identify the student–project matching tactics. The method enables the algorithm to learn from experiments and interactions with the environment by simulating the assignment procedure as an RL problem. Based on encouragements connected to project results and student happiness, QLA directs the agent to investigate and take advantage of activities. By independently assessing the worth of states and the benefit of certain actions, the Dueling Deep Q-Net-work structure improves learning stability. This combination allows for successful adaptability to various and changing student–project circumstances as well as ongoing policy improvement. The QLA-D2QNet algorithm will be described in detail step by step in the

subsequent sections in the context of its components, functionality, reward scheme, and how it enhances dynamic student-project matching efficiency. Algorithm 1 shows the pseudocode for QLA-D2QNet.

Algorithm 1 QLA-D2QNet

```

Initialize replay memory  $D$  with capacity  $N = 100000$ 
Initialize dueling DQN  $Q$  with random weights  $\theta$ 
Initialize target dueling DQN  $Q_{\text{target}}$  with  $\theta_{\text{target}} \leftarrow \theta$ 

Initialize Q-learning-based tuning:
    Learning rate  $\alpha = 0.0005$ 
    Discount factor  $\gamma = 0.99$ 
    Exploration rate  $\epsilon = 1.0$ 
    Minimum  $\epsilon_{\text{min}} = 0.01$ 
    Decay rate  $\epsilon_{\text{decay}} = 0.995$ 

Set batch_size = 64
Set target_update_frequency  $C = 1000$ 
Set max_episodes = 500
Set max_timesteps = 200
For episode = 1 to 500 do:
    Initialize environment and get initial state  $s_0$ 
    total_reward = 0
    For t = 1 to 200 do:
        With probability  $\epsilon = 1.0$  (initially), select random action  $a_t$ 
        Otherwise,  $a_t = \arg\max_a Q(s_t, a; \theta)$ 
        Execute action  $a_t$ , observe reward  $r_t$  and next state  $s_{t+1}$ 
        Store  $(s_t, a_t, r_t, s_{t+1})$  in replay memory  $D$ 
        If len( $D$ ) > batch_size:
            Sample 64 transitions from  $D$ 
            For each  $(s_j, a_j, r_j, s_{j+1})$ :
                If  $s_{j+1}$  is terminal:
                     $y_j = r_j$ 
                Else:
                    Compute Q-values from the target network:
                     $a^* = \arg\max_a Q(s_{j+1}, a; \theta_{\text{target}})$ 
                     $y_j = r_j + \gamma * Q_{\text{target}}(s_{j+1}, a^*; \theta_{\text{target}})$ 
                    → e.g.,  $y_j = 1.5 + 0.99 \times 5.2 = 6.648$ 
                Compute loss:
                 $L = \text{Mean Squared Error between } y_j \text{ and } Q(s_j, a_j; \theta)$ 
                → e.g.,  $L = (6.648 - 6.2)^2 = 0.2016$ 
            Perform gradient descent on  $\theta$  using  $\alpha = 0.0005$ 
        Every 1000 steps:
            Update target network weights:  $\theta_{\text{target}} \leftarrow \theta$ 
        Update  $\epsilon$ :  $\epsilon = \max(\epsilon \times \epsilon_{\text{decay}}, \epsilon_{\text{min}})$ 
        →  $\epsilon = \max(1.0 \times 0.995, 0.01) = 0.995$ 
    total_reward +=  $r_t$ 
     $s_t \leftarrow s_{t+1}$ 
    If done: break
    Use Q-learning reward trends to tune  $\alpha$  or  $\gamma$  if needed:
        If avg_reward < threshold, decrease  $\alpha$ :  $\alpha \leftarrow \alpha \times 0.9 = 0.00045$ 
        Else, slightly increase  $\gamma$ :  $\gamma \leftarrow \min(\gamma + 0.005, 0.999) = 0.995$ 
Return trained Dueling DQN  $Q(\theta)$ 

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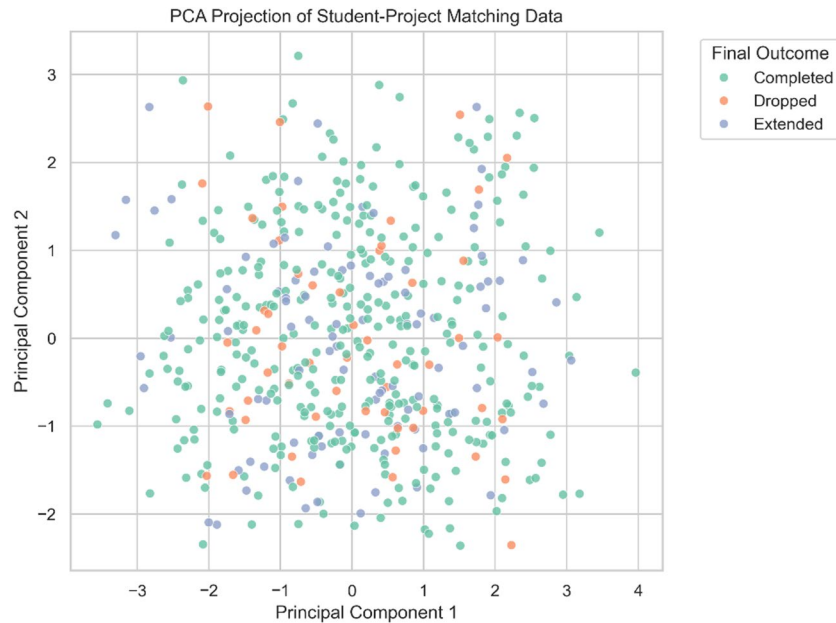


Fig. 5 PCA scatter plot showing project outcomes by student feature distribution

3.4.1 Q-Learning algorithm

QLA increasingly defines valid student-project assignment policies by trial-and-error feedback interactions with the QLA through the learning of action values as a function of reward input so that it can make adaptive decisions to optimize matching performance and learning satisfaction in the long term.

Q-Learning uses a Markov Decision Process (MDP) architecture to continuously learn the best assignment techniques through trial and error in the setting of dynamic project-student pairing. It gathers information from the environment, and a reward measures how well a pupil performs on a project. The primary objective is to maximize the discounted return of rewards, as defined in Eq. 7.

$$H_s = \sum_{\tau=1}^S \gamma^{\tau-1} q(t_s, b_s) \quad (7)$$

While $q(t_s, b_s)$ is the reward obtained after doing action b_s in state t_s , H_s is the return from the time step s , and $\gamma \in [0,1]$ is the discount factor that gives priority to immediate rewards over distant ones.

When performing an action b_s in state t_s and thereafter adhering to the rules π , R , F represents reward and mapping function respectively, Q-Learning estimates the expected return using the action-value function (Q-function). It is explained as follows in Eq. (8):

$$R_{\pi}(t_s, b_s) = F_{\pi}[H_s | t_s, b_s] \quad (8)$$

Actions stand in for possible assignments, states for student profiles and project needs, and motivations for results like project success and student happiness in the suggested dynamic matching approach. Finding the best policy that continuously chooses tasks with the highest long-term return is the purpose.

The predicted reward formulation in Eq. (7) may be rewritten sequentially to yield the Bellman equation, which reflects the link between the current state-action quality and future benefits as provided by Eq. (9):

$$R_{\pi}(t_s, b_s) = F_{\pi}[q(t_s, b_s) + \gamma F[R_{\pi}(t_{s+1}, b_{s+1})]] \quad (9)$$

The outer anticipation is determined by the agent's policy (π), whereas the inner assumption is solely determined by the environment's movements. This recursive approach enables the Q-function to be learned in an off-policy way, allowing the agent to learn the optimum Q-values while exploring with an alternate technique.

Q-Learning employs the Temporal Difference (TD) approach for this purpose. The TD update rule updates the Q-values incrementally, as in Eq. (10).

$$R_{\pi}(t_s, b_s) = R(t_s, b_s) + \alpha [q_{s+1} + \gamma \max_{b_{s+1}} R(t_{s+1}, b_{s+1}) - R(t_s, b_s)] \quad (10)$$

The learning speed $\alpha \in [0,1]$ determines how rapidly Q-values are altered during training. A greater value results in quicker updates. The discount factor, γ , affects the relevance of future benefits over immediate ones.

The expression t_{s+1} refers to the reward earned after completing an action in the present state. Finally, the conditional $\max_{b_{s+1}} R(t_{s+1}, b_{s+1})$ reflects the most accurate forecast of the future state based on the agent's existing knowledge. These components work together to help the agent learn and make better judgments over time. Figure 6 shows an example of the Q-table with associated Q-values, which guide the agent's policy updates during the training phase.

For student-project matches, the Q-table is a projection of values for matching a given student profile (state) to a given project. As the agent moves forward, operating in the world through assignments and gaining inducements (e.g., for success, satisfaction, skill match), the Q-values are updated to reflect the goodness of each match. Along the way, the agent comes to select actions that lead to big, long-term rewards, learning in the process optimal dynamic matching strategies. With every cycle of the Q-Learning process, the agent interacts with the world through a sequence of procedures to enhance its decision-making over time:

- Observes present status t_s including student-specific factors like hobbies, talents, and academic background.
- Chooses and conducts an action b_s , signifying the student's assignment to a certain training project.
- Observes the ensuing next state t_{s+1} and any movement or changes to the environment (e.g., project slot updates or remaining student pool).
- Receives an instant reward (q_s) depending on indicators, including student happiness, project relevancy, and learning results.
- Q-Learning is an off-policy method that can assess and enhance the target policy π while adhering to a distinct behavior policy. The greedy policy, $\mu(t) = \operatorname{argmax}_b R(t, b)$ picks the measure with the largest Q-value for an instance.

Following training, the agent employs this greedy strategy to select actions by determining the highest-valued assignment in the Q-table, as shown in Eq. (11):

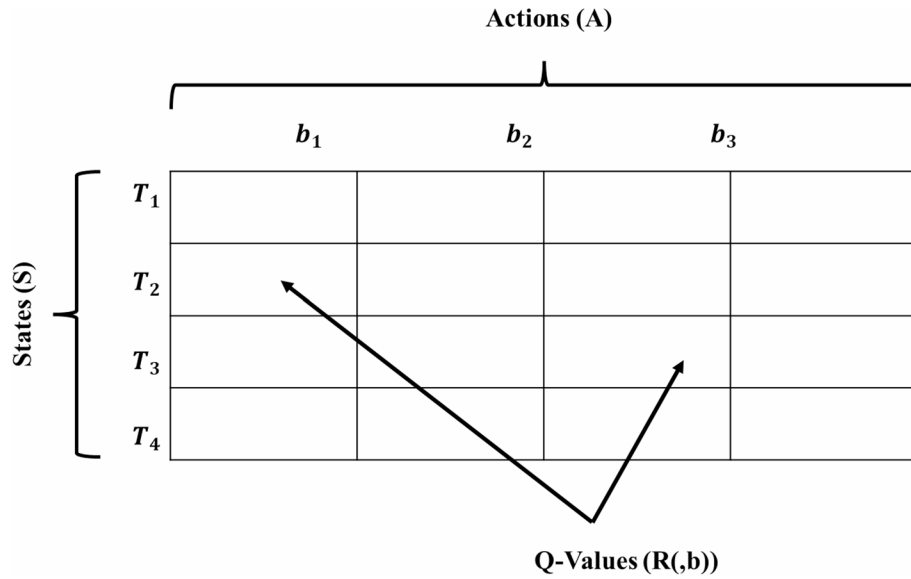


Fig. 6 Q-table and associated Q-values

$$b_s = R[t_s, \mu(t)] R[t_s, \argmax_b R(t, b)] \quad (11)$$

In this case, a is the specified action in the condition t_s , and R is the current phase-action value table. Thus, the agent picks the project assignment that optimizes long-term benefits for the given student's demographic.

The optimum policy (π^*) is accomplished when the agent constantly chooses the best feasible actions throughout all states. The optimum Q-function R^* can be recursively defined Eq. (12):

$$R^* = q_j + \gamma \max_{b'} F[R^*(t_{s+1}, b')] \quad (12)$$

where, R^* represents the optimal expected reward, q_j is the immediate reward for action j , γ is the discount factor for future rewards, b' denotes possible next actions, and $F[\cdot]$ is the mapping function that computes rewards from the next state t_{s+1} and action b' . This equation recursively selects the action leading to the maximum cumulative reward.

This formulation assures that the agent's decision-making is continually refined to maximize the projected cumulative reward. The dynamic matching architecture allows for the discovery of highly successful student-project pairings through interaction with the environment, eventually improving the quality and relevance of training project assignments.

3.4.2 DQN

To improve the Q-Learning process by distinguishing between value and advantage functions. This design enhances learning stability and decision accuracy, allowing for more effective and dependable dynamic student-project pairing in creative training settings.

In the dynamic matching approach for college students' creative training initiatives, the Q-learning algorithm uses a Q-table to store and update the predicted reward values for each state-action combination. However, when the variety of learners, projects, and

qualities grows, the state space becomes intractable. To remedy this, Deep DQN is used to estimate Q-values using DL techniques. DQN substitutes the Q-table with an AI system that forecasts the Q-value for a particular state-action combination. The network is trained with a loss mechanism based on the TD error given by Eq. (13):

$$K(x) = F \left[\left(Q_{s+1} + \gamma \max_b R(T_{s+1}, b) - R(T_s, B_s, x) \right)^2 \right] \quad (13)$$

Here, $K(x)$ represents a loss or error measure used to update the model. Q_{s+1} is the predicted value at the next state, $F[\cdot]$ denotes a function or mapping $\max_b R(T_{s+1}, b)$ is the maximum expected reward over possible actions at the next state. $R(T_s, B_s, x)$ is the network's anticipated Q-value with weights x , whereas γ is the reduction ratio. To reduce loss, the network is trained by dynamic gradient descent, allowing the agent to learn optimum student-project matching rules even in high-dimensional domains. This method improves flexibility and efficiency in comparing judgments with experience.

3.4.3 D2QNet

This structure divides the Q-value into two elements: the state value function $U^\pi(t)$, which depends on the total worth of being in a particular state (e.g., a student's aptitude profile), and the benefits function $B^\pi(t, b)$, which captures the further advantage of providing a specific project b in state t .

The connection is defined as follows: Eq. (14):

$$B^\pi(t, b) = R^\pi(t, b) - U^\pi(t) \quad (14)$$

By breaking down the Q-function in this manner, the network may learn more efficiently learn which states are valued regardless of action and which events generate the most in valued regions. This design is especially useful in situations such as student-project comparison, where some students may have generally similar profiles, making action-specific judgment less necessary. The Dueling DQN thereby increases learning stability and speeds convergence to optimal matching techniques.

Figure 7 shows the structure of the Dueling DQN model, which is meant to effectively improve decision-making in complicated situations such as student-project matching. This system draws experience samples from the memory buffer, which include student identities, assigned actions (projects), resultant states, and awards, and passes them via the Dueling DQN's input layer.

The model is divided into two sources: one predicting the state value $U(t; \theta, \beta, \beta)$ and the other calculating the action advantage $B(t, b; \theta, \alpha)$. θ indicates common extracted feature factors, whereas α and β are unique to each stream.

The total output Q-value is computed as in Eq. (15):

$$R(t, b; \theta, \alpha, \beta) = U(t; \theta, \beta) + B(t, b; \theta, \alpha) \quad (15)$$

However, because the decomposition is non-unique, this straight combination may be unstable. To solve this, the advantage term is normalized by eliminating the highest (or average) benefits across actions, yielding the refined Q-value, shown by Eq. (16):

$$R(t, b; \theta, \alpha, \beta) = U(t; \theta, \beta) + B(t, b; \theta, \alpha) - \max_{\alpha' \in B} (t, b'; \theta, \alpha) \quad (16)$$

where:

- $U(t)$ represents the value of being in state t ,
- $\max_{\alpha' \in B}$ means yield the maximum value over all possible choices of α' in the set B .
- $B(t, b)$ represents the advantage of choosing action b in state t ,
- $R(t, b)$ represents the overall action-value function,
- θ are shared network parameters, while α and β are parameters for advantage and value streams, respectively.

Equation 16 ensures that the network better distinguishes between valuable states even when actions have similar outcomes, improving stability and convergence.

The D2QNet architecture is shown in Fig. 7, where a memory buffer guides reinforcement learning's interactions with the environment through states, actions, and rewards. Convolutional layers are used by the duelling DQN to extract features, which are then divided into value (U) and advantage (B) streams. The output layer combines these streams to calculate the final action-value function, $R(t, b)$.

The training process of the proposed QLA-D2QNet is represented as follows:

1. *Experience replay*: Transitions (s_t, a_t, r_t, s_{t+1}) are stored in a replay buffer of size $N = 100,000$.
2. *Mini-batch sampling*: At each iteration, a random mini-batch of 64 samples is drawn from the buffer.
3. *Target network update*: A target duelling DQN network θ^- is updated every 1000 steps to stabilize training.
4. *Loss function*: The mean-squared error loss is used

$$L(\theta) = E_{(s_t, a_t, r_t, s_{t+1})} \left[\left(r + \gamma \max_{a'} Q(s', a'; \theta^-) - Q(s, a, \theta) \right)^2 \right]$$

where $\gamma = 0.99$ is the discount factor.

Here, $L(\theta)$ is the mean squared temporal difference (TD) loss used to train the Q-network. (s_t, a_t, r_t, s_{t+1}) are the state, action, reward, and next state samples from experience; γ is the discount factor, $Q(s', a'; \theta^-)$ is the target Q-value from the target network, and $Q(s, a, \theta)$ is the current Q-value prediction. This loss measures the difference between predicted and target Q-values to update network parameters θ .

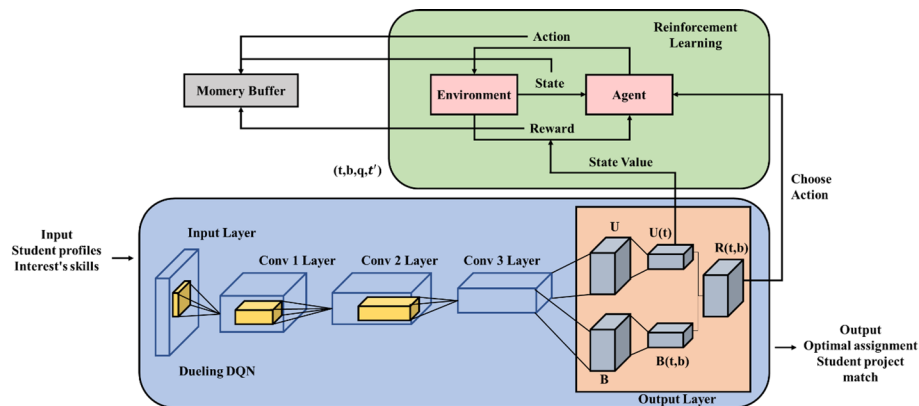


Fig. 7 Architecture of D2QNet

5. *Exploration strategy*: Q-learning with ϵ -greedy policy is employed, where ϵ decays linearly from 1.0 to 0.1.

4 Reward function design

The reward function incorporates several goals, including project success, domain relevance, learning style compatibility, skill-interest fit, and student-project matching. To facilitate steady policy learning, these elements are integrated into a weighted reward formulation.

$$R = w_1.SkillInterestFit + w_2.Domain\ Relevance \\ + w_3.LearningStyleFit + w_4.ProjectSuccess$$

In the equation, the weights w_1 , w_2 , w_3 , and w_4 are empirically tuned as per the requirement. This formulation balances cognitive, technical, and outcome-driven aspects of project matching.

A model termed Dueling DQN makes use of the average advantage to guarantee stability and seamless learning in practical applications. This framework enables the model to efficiently assess student profiles and modify matching choices in response to project fit variations, enhancing the efficacy and caliber of learning in dynamic learning environments. The QLA-D2QNet model is a dynamic student-project matching system that combines Q-Learning and Dueling Deep Q-Network techniques, improving learning stability and enhancing convergence speed in dynamic environments. It outperforms single Q-Learning or Dueling DQN techniques in comparison tests. Table 2 shows the hyperparameters for QLA-D2QNet.

Based on empirical testing and best practices for reinforcement learning, QLA-D2QNet's hyperparameters were chosen to balance learning efficiency, stability, and convergence speed. The learning rate provides constant gradient updates, while the replay memory capacity ensures a large buffer for a variety of experiences. By prioritizing long-term incentives, the discount factor is in line with the cumulative performance of student projects. The hyperparameters (e.g., replay memory $N=100000$, learning rate $\alpha=0.0005$) were selected based on widely adopted DRL literature and preliminary experiments, ensuring stable convergence and efficient learning. A smaller memory or higher learning rate caused instability, while larger values slowed training. Within reasonable ranges, the model maintained robust performance, confirming the appropriateness of the chosen settings. The goal network update frequency ($C=1000$) and batch size (64) maximize computational efficiency and training stability. According to a hyperparameter sensitivity analysis, QLA-D2QNet continuously maintained excellent matching accuracy and reward trends over acceptable fluctuations, confirming the stability of the selected settings.

5 Result

The experiments were conducted using Python 3.9. The suggested approach is evaluated using performance metrics such as Matching Accuracy, Student Satisfaction, Project Outcome, and Learning Outcome Impact. Both the QLA baseline model and the proposed QLA-D2QNet method were trained and evaluated on a student-project matching dataset.

The correspondence between team size and innovation index for research, survey, simulation, and prototype projects is presented in the bubble chart of Fig. 8. Two to five team members are plotted on the y-axis, while the innovation index of 0.45 to 0.95 is plotted on the x-axis. The color of each bubble sets the type of project, and the size of each bubble sets the amount of the prize, which varies from 0.56 to 0.96. Interestingly, the maximum reward values were obtained by Research and Prototype projects that clustered between innovation indices of 0.8 to 0.95 and team sizes of 3 to 4. The visualization reveals that higher innovation and balanced team sizes lead to better project outcomes, while survey and simulation projects are more common and offer fewer awards, highlighting the importance of reward-driven results in project matching.

The frequency of students' learning styles used by a college is illustrated using a heatmap in Fig. 9. The four types of projects are prototype, research, simulation, and survey, from left to right in the horizontal axis, and the three most common learning styles, visual, kinesthetic, and auditory, are from top to bottom in the vertical axis. The ratio of students with a specific learning style on a specific type of project is viewed through the percentage values in each cell. Notably, kinesthetic and auditory students are more evenly dispersed among research, simulation, and survey kinds, but visual learners have a substantial predilection for simulation projects (32.7%). Prototype projects consistently have low engagement across all learning styles, highlighting areas of mismatch and concentration. To improve student engagement and academic results, customized project assignment methodologies should be implemented that suit students' preferred learning methods.

A boxplot is used in Fig. 10 to show the distribution of reward values across different skill match scores (0–4) and innovation index quartiles (Low, Mid, High, and Very High). The reward value, which runs from 0.5 to 1.0, is represented by the y-axis. In the low-performing and medium innovative quarters (blue shades), skill match score 0 is notable for having greater median payouts (~ 0.78 – 0.80), whereas the high and very high quarters (reddish colors) have significantly lower median rewards (~ 0.72). The reward amounts are still evenly distributed across quartiles at skill match score 1, with the low and mid quartiles once more reaching medians over 0.75. The reward values in the high and very high innovation quartiles decrease when the skill match score rises to 3 and 4, with medians as low as ~ 0.68 – 0.70 , suggesting potential mismatches.

Table 2 Hyperparameter table of QLA-D2QNet

Hyperparameter	Symbol	Value
Replay memory capacity	N	100,000
Learning rate	α	0.0005 (adaptive)
Discount factor	γ	0.99 (adaptive)
Initial exploration rate	ϵ	1.0
Minimum exploration rate	$\epsilon_{\min}$	0.01
Exploration decay rate	ϵ_{decay}	0.995
Batch size	–	64
Target network update Freq.	C	1000 steps
Max episodes	–	500
Max timesteps per episode	–	200
Loss function	–	MSE (L2 Loss)
Optimizer	–	Adam
Architecture	–	Dueling DQN
Q-Tuning trigger threshold	–	avg_reward < 50

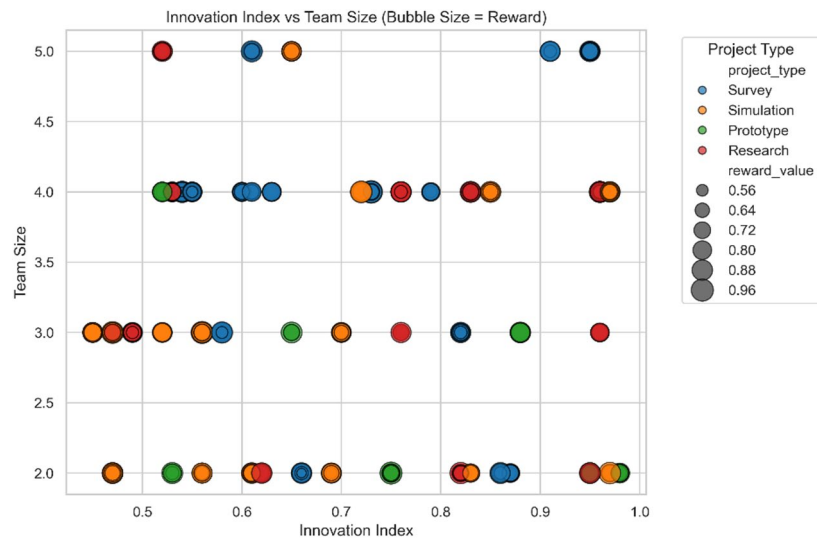


Fig. 8 Relationship between innovation index, team size, and project type with reward scaling

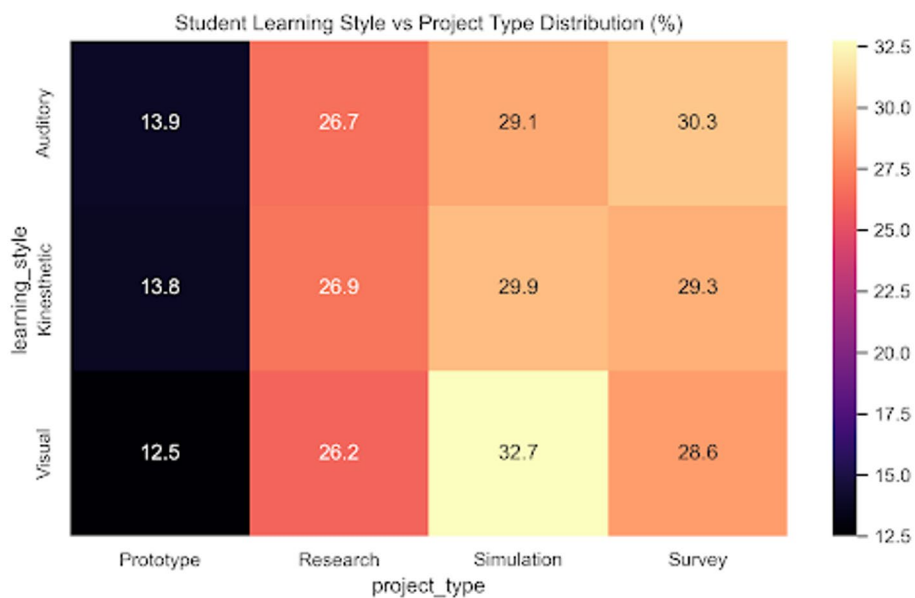


Fig. 9 Heatmap of student learning styles versus project type distribution (%)

5.1 Hyperparameter sensitivity analysis

The learning rate (α) and replay memory capacity (N) are two important hyperparameters that were performed a sensitivity analysis of to make sure the suggested QLA-D2QNet model is resilient. The differences in performance under various parameter settings are compiled in Table 3. Variations in memory capacity had a moderate impact on training efficiency but did not impair overall performance, according to the results, while very small or large learning rates resulted in sluggish convergence or instability. The selected values ($\alpha=0.0005$, $N=100,000$) demonstrated a balanced combination of fast convergence, high average reward, and stable learning, indicating their appropriateness for the suggested model.

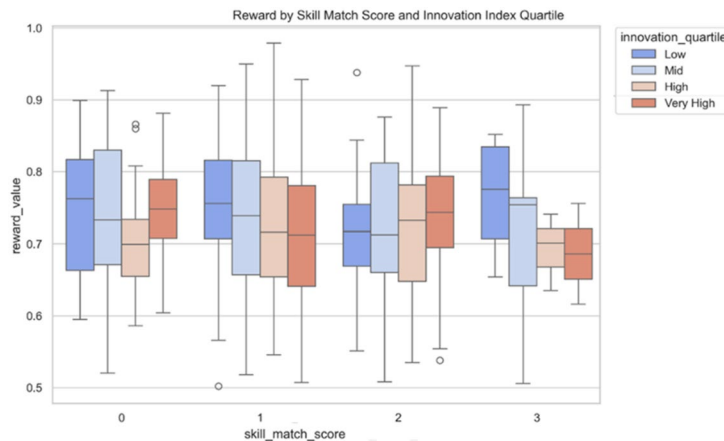


Fig. 10 Boxplot of reward distribution by skill match score and innovation index quartile

Table 3 Sensitivity analysis of key hyperparameters in QLA-D2QNet

Learning rate (α)	Replay memory (N)	Average reward	Convergence speed	Stability
0.0001	50,000	Moderate	Slow	Stable
0.0005 (<i>chosen</i>)	100,000	High	Fast	Stable
0.001	100,000	Fluctuating	Fast	Unstable
0.0005	150,000	High	Moderate	Stable

5.2 Performance analysis of QLA-D2QNet

Matching accuracy relates to how effectively student-project assignments correspond to students' abilities, interests, learning styles, and project significance. In this goal, RL dynamically increases matching accuracy by learning from criticism and optimizing subsequently assigned selections. Figure 11; Table 4 show that QLA-D2QNet outperforms QLA. The suggested model outperforms all other models tested. Skill-Interest Fit improved from 68.40% with QLA to 89.10% with QLA-D2QNet, showing a greater match between students' talents and project requirements. Area-Relevance Alignment also increased significantly, from 72.00% to 90.50%, indicating more precision in allocating students to projects within their academic or technical area. Furthermore, the Learning Style Fit increased significantly from 59.70 to 85.30%, indicating that the suggested methodology is more effective at tailoring project assignments to different needs for learning.

Student satisfaction is the extent to which project assignments appeal to students' interests, goals, and learning styles. To meet this goal, RL increases satisfaction with ongoing feedback from students and results in learning to make project assignments more individualized to meet expectations and incentives. Figure 12; Table 5 compare student satisfaction and engagement metrics between QLA and the proposed QLA-D2QNet model. The satisfaction score increased from 3.2 QLA to 4.6 QLA-D2QNet, suggesting a better overall student experience. The Project Engagement Rate also rose dramatically, from 65.00 to 92.20%, indicating higher student participation in given tasks. The Dropout Rate has dropped significantly, from 21.30% with QLA to just 6.50% with QLA-D2QNet, indicating more effective and engaging project assignments. Additionally, the Mentor Feedback Score increased from 3.5 to 4.7, indicating a better mentor view of pupil achievement and fit.

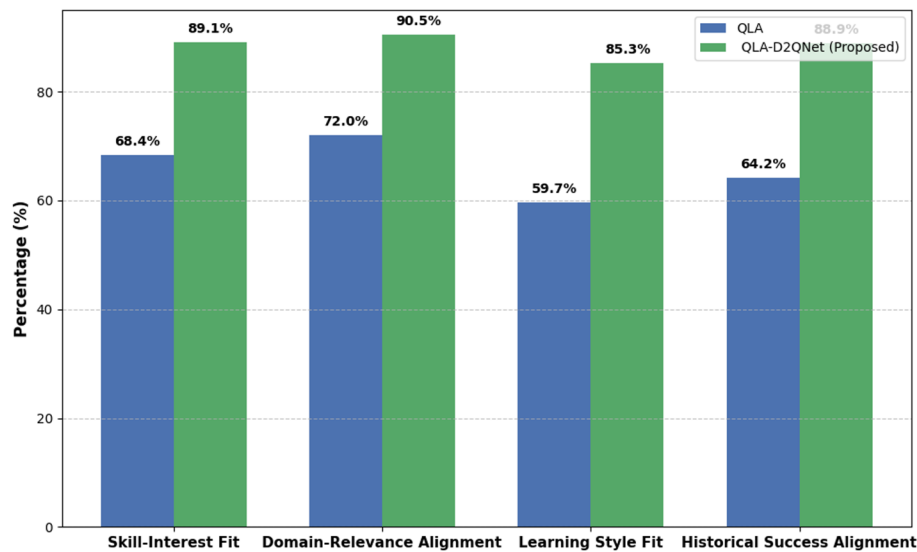


Fig. 11 Performance of QLA and QLA-D2QNet in matching effectiveness metrics

Table 4 Performance comparison of QLA and proposed QLA-D2QNet model on matching accuracy sub-metrics

Matching accuracy (%)	QLA	QLA-D2QNet (Proposed)
Skill-Interest fit	68.40	89.10
Domain-Relevance alignment	72.00	90.50
Learning style fit	59.70	85.30
Historical project success alignment	64.20	88.90

The suggested QLA-D2QNet model beats the traditional QLA in project outcomes. The project completion rate increased from 71.80 to 94.00%, and the innovation grade improved from 6.3 to 8.7, suggesting more inventiveness and output quality. Peer assessment scores increased from 3.4 to 4.5, indicating improved teamwork. The prototype or output submission rate also increased significantly, from 54.60 to 87.30%. Furthermore, the average matching time per batch decreased from 18.4 to 4.2 s, demonstrating the model’s increased efficiency in producing faster and more successful student-project assignments. Figure 13 compares project outcomes and efficiency metrics, showing improved performance of the proposed QLA-D2QNet over standard QLA.

Learning outcome effect assesses students’ skill development, academic alignment, and assurance obtained from projects. In this goal, RL improves the learning effect by dynamically assigning tasks that are educationally effective and customized to student development needs based on continuous feedback. When compared to QLA, the QLA-D2QNet model produces much better learning outcomes. Course integration success increased from 54.7% to 85.9%, indicating a greater match with academic objectives. Student confidence scores increased from 3.1 to 4.6, indicating more self-assurance. Furthermore, the re-application rate for future projects went from 43% to 77.8%, indicating that students are more interested and satisfied with continuing their innovative training initiatives. Figure 14 illustrates learning outcome metrics highlighting the proposed QLA-D2QNet’s improvement in skill acquisition, confidence, and re-use rate.

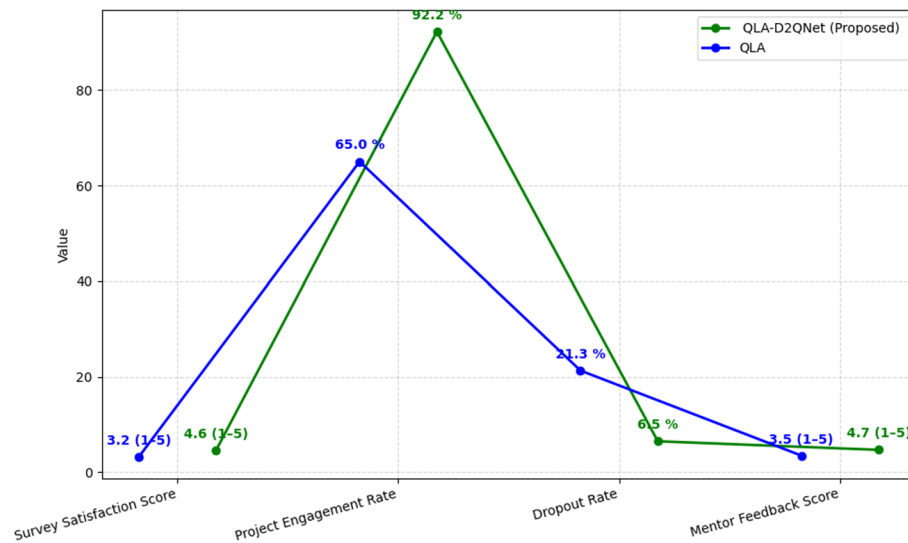


Fig. 12 Comparison of student experience metrics for QLA and QLA-D2QNet models

Table 5 Comparison of student satisfaction and engagement metrics between QLA and QLA-D2QNet

Student satisfaction	QLA	QLA-D2QNet (Proposed)
Survey satisfaction score (1–5)	3.2	4.6
Project engagement rate (%)	65.00	92.20
Dropout rate (%)	21.30	6.50
Mentor feedback score (1–5)	3.5	4.7

5.3 Comparison of student-project matching model

The matching accuracy of the suggested QLA-D2QNet model was tested with a baseline (MeanShift – K-Nearest Neighbors) MeanShift-KNN [28] method, which is a clustering algorithm and a classification technique that identifies dense regions (modes) in the feature space to group similar data points to assess its efficacy. The QLA-D2QNet outperformed the baseline accuracy of 94.11% with an accuracy of 96.47%, as indicated in Table 6. This shows that by dynamically learning optimal assignment policies, the reinforcement learning-based model can more accurately and successfully capture student-project compatibility, resulting in more accurate and efficient project allocation decisions.

Figure 15 illustrates the accuracy improvement of QLA-D2QNet over the baseline MeanShift-KNN [28]. The shaded region highlights its potential variability in accuracy. QLA-D2QNet achieves the highest accuracy, demonstrating its superior performance in student-project matching.

5.4 Statistical significance analysis

Statistical significance tests were performed to validate the effectiveness of the suggested QLA-D2QNet. A one-way ANOVA was conducted across all metrics to examine consistency, and an independent sample t-test ($n = 30$ runs) was used to investigate differences between QLA and QLA-D2QNet on key metrics.

According to the findings, which are compiled in Table 7, QLA-D2QNet's enhancements are statistically significant ($p < 0.01$) for every performance metric. This

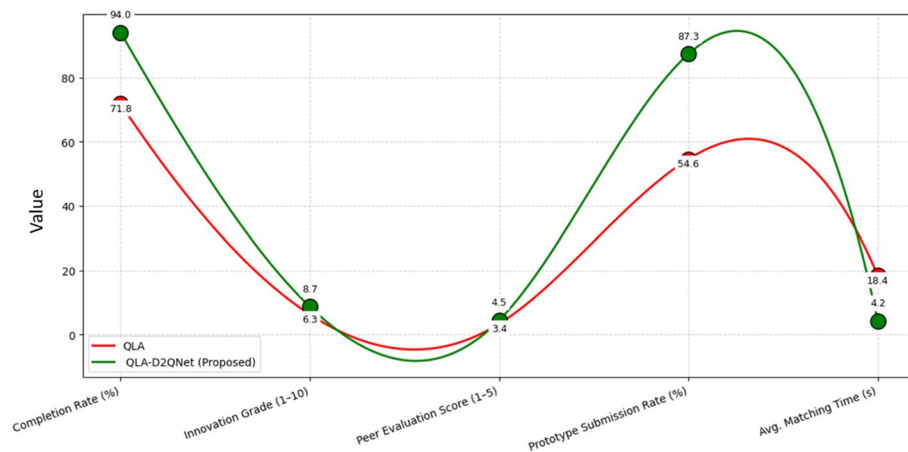


Fig. 13 Evaluation of completion, innovation, and efficiency between matching models

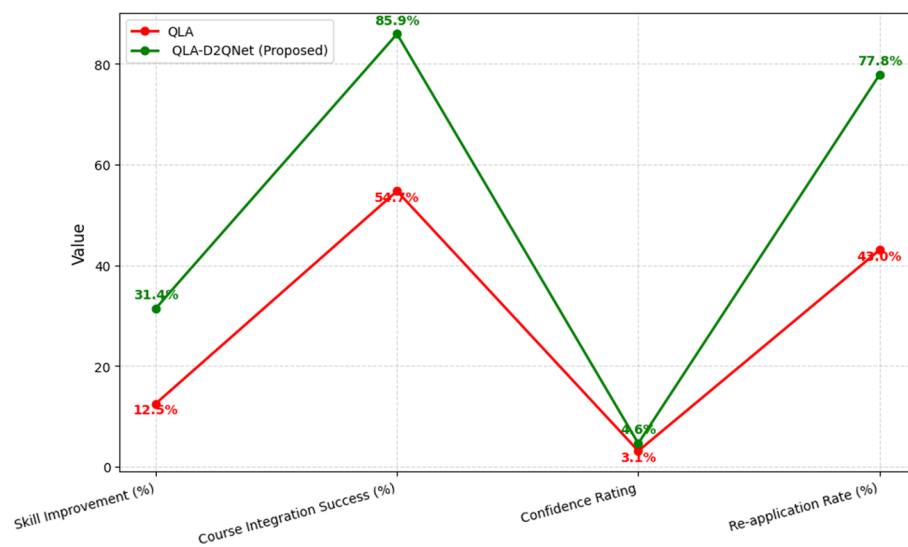


Fig. 14 Effectiveness of QLA-D2QNet in enhancing student outcomes

Table 6 Accuracy comparison of baseline and proposed models

Models	Accuracy (%)
MeanShift-KNN [28]	94.11
QLA-D2QNet [Proposed]	96.47

demonstrates that the noted enhancements are not coincidental but rather represent a strong benefit of the suggested paradigm. Table 7 summarizes the results of statistical significance obtained for the proposed method.

Additionally, the statistical significance analysis demonstrates how strong the suggested framework is. Both the independent sample t-test and the one-way ANOVA confirmed $p < 0.01$ across all parameters, demonstrating the consistently significant gains in matching accuracy, student satisfaction, and project outcomes that QLA-D2QNet accomplished. This suggests that the observed increases represent actual model performance advantages rather than being the result of chance variance. The validity of

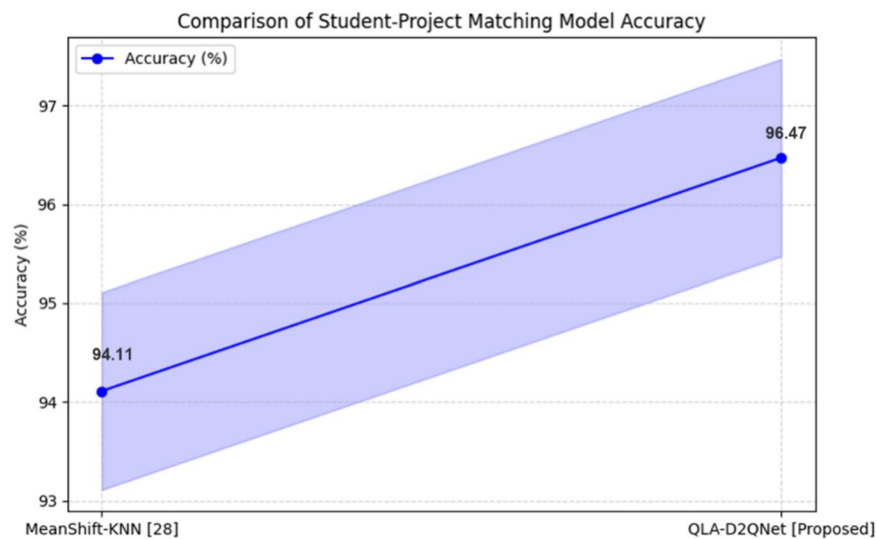


Fig. 15 Graphical representation of the accuracy of student-project matching models

Table 7 Statistical significance outcomes of the QLA-D2QNet approach

Metric	QLA (Mean $\pm$ SD)	QLA-D2QNet (Mean $\pm$ SD)	t-test <i>p</i> -value	ANOVA <i>p</i> -value
Skill-Interest fit (%)	68.4 $\pm$ 2.1	89.1 $\pm$ 1.8	< 0.001	< 0.001
Domain-Relevance alignment (%)	72.0 $\pm$ 1.9	90.5 $\pm$ 2.0	< 0.001	< 0.001
Project completion rate (%)	71.8 $\pm$ 2.4	94.0 $\pm$ 1.6	< 0.001	< 0.001
Student satisfaction score (1–5)	3.2 $\pm$ 0.3	4.6 $\pm$ 0.2	< 0.001	< 0.001

QLA-D2QNet as a dependable and scalable solution for dynamic student–project matching in creative training contexts is reinforced by these findings.

5.5 Ablation study

Q-Learning alone, Dueling DQN alone, and the hybrid QLA-D2QNet fusion were compared in an ablation research to better demonstrate the contributions of each element in the suggested QLA-D2QNet model that was represented in Table 8. In the student-project matching task, this research assesses the effects of each component on matching accuracy, skill-interest alignment, and learning style fit. By separating the effects of each component, we can show how Q-Learning, Dueling DQN, and the adaptive hyperparameter tuning process add value.

Every element contributes to the total performance, as the ablation results make clear. The simplest results are obtained by standalone Q-Learning because of its instability in high-dimensional state spaces. Though it doesn't adaptively adjust learning settings, dueling DQN increases stability and learning accuracy. All measures show that the fusion mechanism and adaptive tuning greatly improve matching accuracy, skill-interest alignment, and learning style fit, with the whole QLA-D2QNet model achieving the best results. The originality and usefulness of our suggested strategy above single-method baselines are highlighted by these results.

Table 8 Ablation study of QLA-D2QNet components

Method	Skill-Interest fit (%)	Domain-Relevance alignment (%)	Learning style fit (%)
Q-Learning	68.4	70.1	59.7
Dueling DQN	75.2	78.5	67.4
QLA-D2QNet [Proposed]	89.1	90.5	85.3

5.6 Discussion

The suggested RL-based approach boosts learning outcomes, aligns students with projects more effectively, and dynamically adjusts to preferences and feedback on project success. By using deep reinforcement learning to create successful teams based on behavioral data, the research [19] offers a persona-based approach to student grouping that challenges learning results. Furthermore, research [18] investigates AI-based creative exercises; reliable performance predictions were done using SVM, supporting the results, which demonstrate lower critical thinking and problem-solving skills. In every important area, the suggested QLA-D2QNet model dramatically enhanced student-project matching performance. With Domain-Relevance Alignment at 90.50% and Skill-Interest Fit at 89.10%, it produced exceptional Matching Accuracy results. The Project Engagement Rate increased to 92.20%, while the Survey Satisfaction Score increased to 4.6 in terms of student satisfaction. Higher innovation and delivery quality were demonstrated by the model's 94.00% Project Completion Rate and 87.30% Prototype Submission Rate for Project Outcome. Additionally, learning outcomes were improved; the Re-application Rate increased to 77.8% and the Course Integration Success peaked at 85.9%. These enhancements show QLA-D2QNet offers a more flexible, effective, and customized method of project assignment that improves educational impact, increases engagement, and maintains student interest. The interpretability of QLA-D2QNet is ensured through PCA feature contributions and Dueling DQN action-value decomposition, allowing educators to understand why specific student-project matches are recommended. All data are anonymized to protect student privacy, and the model serves as a decision-support tool, preserving human oversight to maintain fairness and ethical allocation.

6 Conclusion

The proposed resource-dependent approach with the usage of RL enhances student performance, nurtures skills, and allows for making flexible and data-driven decisions. The suggested QLA-D2QNet model symbolically overcomes the limits of fixed-point student-project assignments by deploying RL to dynamically match. The system effectively encompasses the features of the students and project by extracting and encapsulating the features via the use of feature extraction techniques, PCA, and normalization as part of the preprocessing earlier. The model recorded high scores of Skill Interests Fit at 89.10, Survey Satisfaction score at 4.6, Project completion rate at 94.00, and Skill Improvement score at 31.40. The approach that integrates QLA-D2QNet allows adapting the corresponding policies on the basis of continuous feedback, which is scalable, effective, and accurate in the educational spheres.

6.1 Limitations and future scope

The model could be affected by the quality and variety of the input data because of its generalizability. Future research can explore better customization and adaptation to

dynamic educational settings by researching hybrid RL strategies, incorporating real-time emotional or behavioral analysis, and extending the system into cross-disciplinary or multi-institutional environments to apply it more broadly.

Author contributions

All authors contributed to the study conception and design. Data collection and analysis were performed by Xiao Ju and Yumo Ding. The first draft of the manuscript was written by Xiao Ju and Yumo Ding. All authors read and approved the final manuscript.

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All data generated or analysed during this study are included in this article.

Declarations

Ethics approval and consent to participate

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Consent for publication

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Competing interests

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